

GIS Features User Guide

1. Map Layer Icon Overview

The Map Layer button () provides complete control over map display layers, allowing you to customize your mapping experience with multiple data sources and visualizations.

2. Available Map Layers

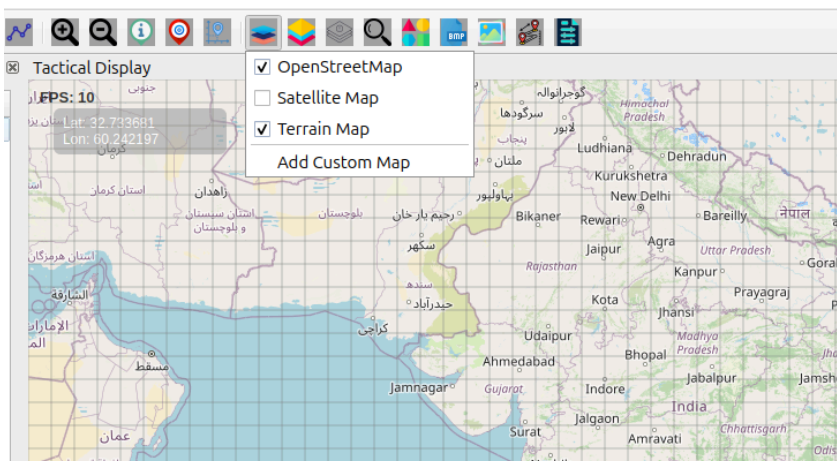
2.1. Built-in Layers

- OpenStreetMap - Standard street and road maps
- Satellite Map - High-resolution aerial imagery
- Terrain Map - Topographic and elevation data

3. How to Use Map Layers

3.1. Accessing Layer Controls

- Click the Map Layer icon () on the Design Toolbar
- A dropdown menu appears showing all available layers
- Check the box to show a layer
- Uncheck to hide a layer
- Multiple layers can be visible simultaneously
- Adjust zoom using + and - buttons or mouse scroll wheel (0-11 zoom levels)



Import GeoJSON Features User Guide

1. Import GeoJSON Icon Overview

The Import GeoJSON button () allows you to add custom geographic data layers to your map from GeoJSON files, enabling you to visualize points, lines, polygons, and other vector data.

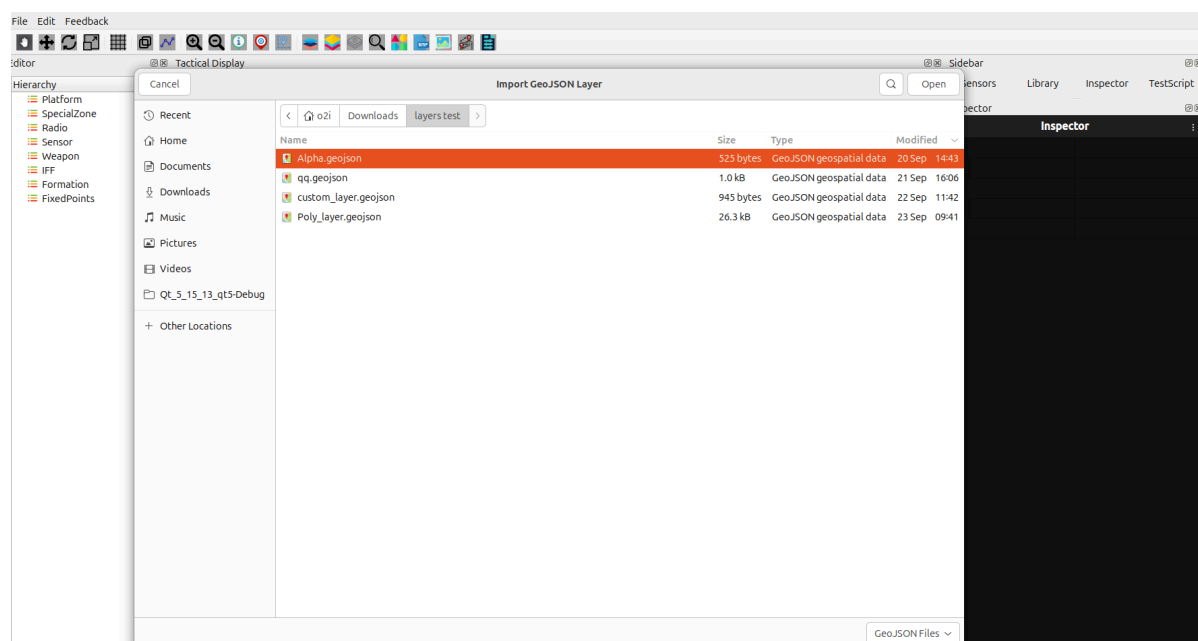
1.1. Supported File Formats

- GeoJSON (.geojson, .json)
- JSON files with GeoJSON structure
- GeoJSONL (Line-delimited GeoJSON)
- TopoJSON (.topojson)

2. How to Import GeoJSON Files

2.1. Step-by-Step Import Process

1. Click the Import GeoJSON icon () on the Design Toolbar
2. File dialog opens - navigate to your GeoJSON file
3. Select the file and click "Open"
4. The layer automatically loads and appears on your map
5. Layer name is derived from the filename



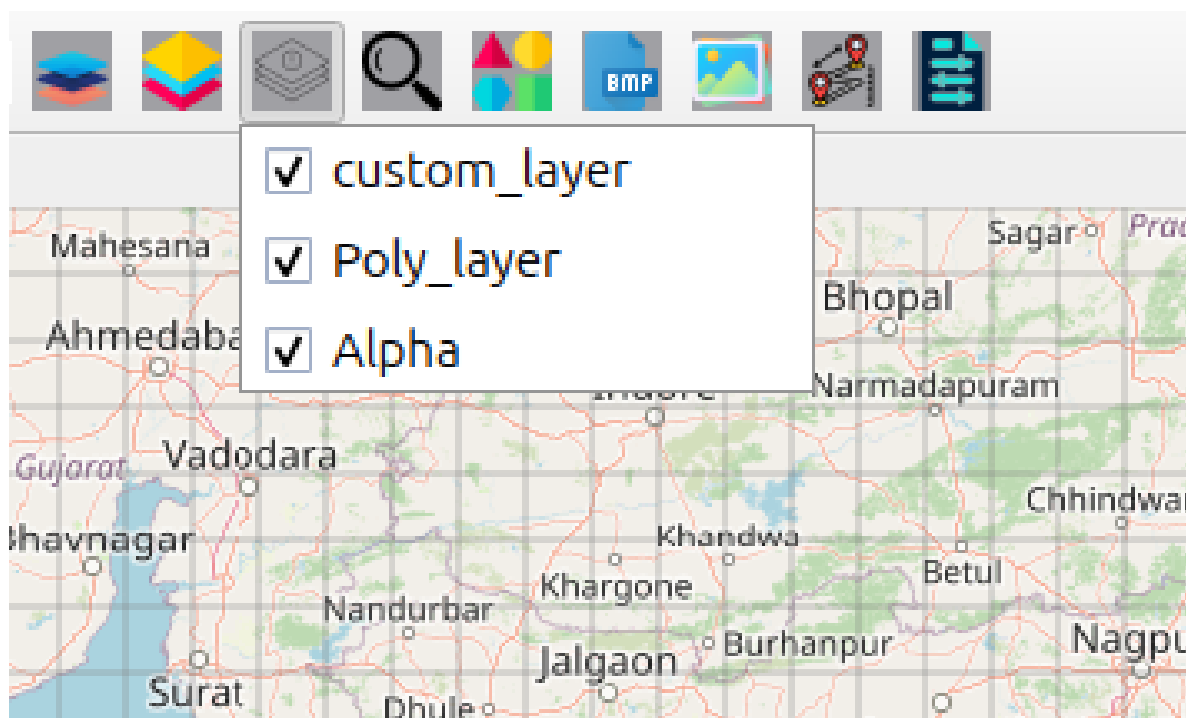
2.2. Automatic Layer Management

- Layer is automatically visible after import
- Added to GeoJSON Layers menu for management
- Appears in the layer visibility controls
- Supports multiple GeoJSON layers simultaneously

3. Managing GeoJSON Layers

3.1. Layer Visibility Control

1. Click the GeoJSON Layers button (layers icon)
2. Dropdown menu shows all imported GeoJSON layers
3. Check the box to show a layer
4. Uncheck to hide a layer
5. Multiple GeoJSON layers can be visible at once



Search Place Features User Guide

1. Search Place Icon Overview

The Search Place button () allows you to quickly find and navigate to any location on the map by name or coordinates. It supports both place name searches and direct latitude/longitude lookup.

2. How to Use Search

2.1. Accessing Search

1. Click the Search Place icon () on the Design Toolbar
2. A search box dropdown appears
3. Enter your search term or coordinates
4. Press Enter or Return to execute search

2.2. Search Methods

1. Place Name Search

Enter any location name:
- "Delhi"

2. Coordinate Search

Enter coordinates in decimal format:
"40.7128, -74.0060" (latitude, longitude)

"lat, lon" format separated by comma

3. Search Results & Actions

3.1. Automatic Map Navigation

- Map automatically centers on search result
- Zoom level adjusts based on place importance:
 - Countries: Zoom level 6
 - States/Regions: Zoom level 7
 - Cities: Zoom level 8
 - Towns/Villages: Zoom level 9

3.2. Visual Indicators

- Red Marker  placed at search location
- Geographic Outline displayed for area features
- Coordinate display in information panel

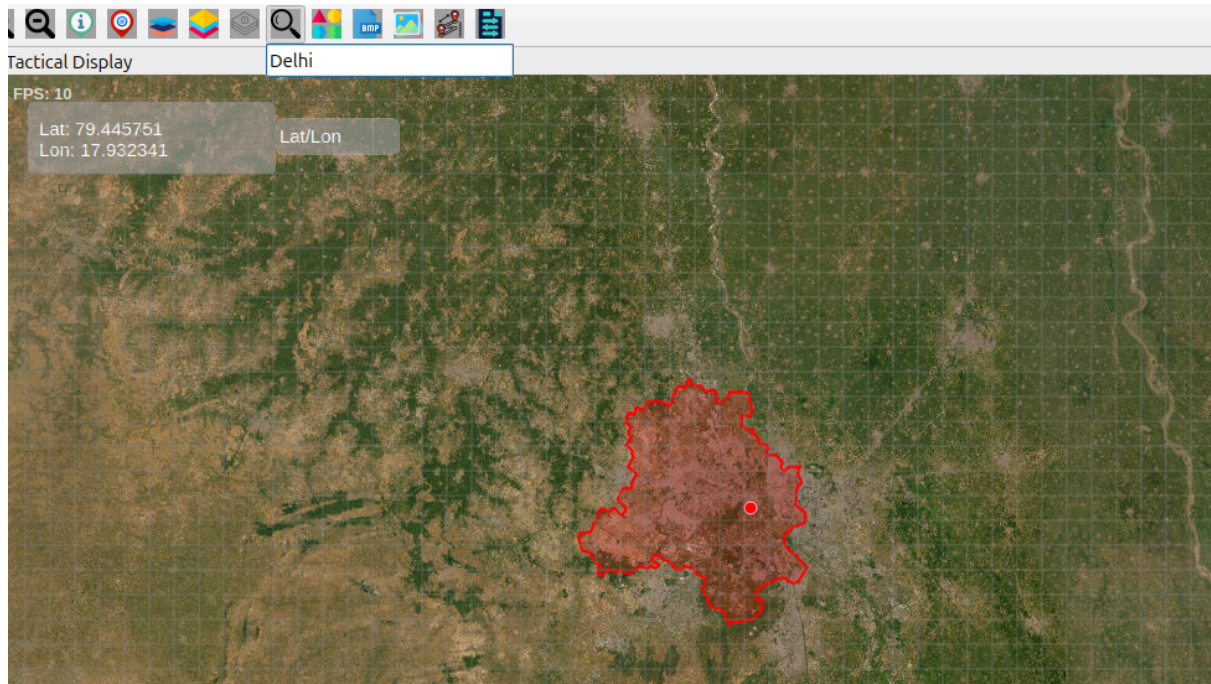


Fig:1. Search by place

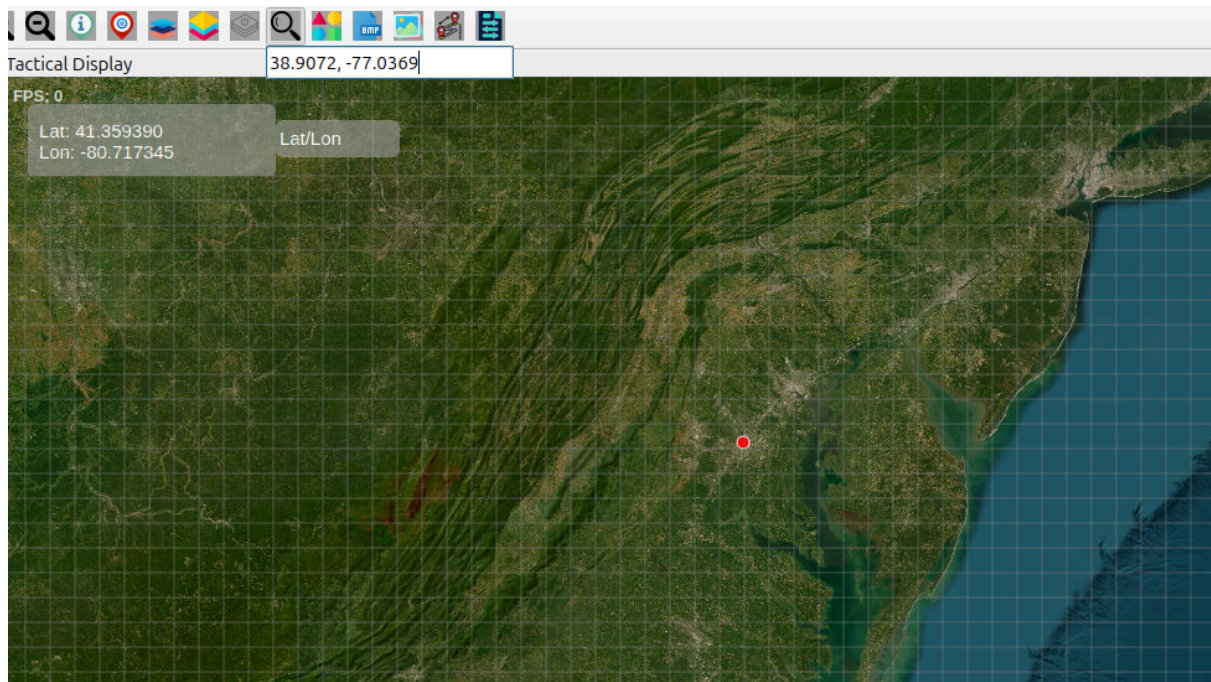


Fig:2. Search by coordinates

Shape Drawing Features User Guide

1. Shape Icon Overview

The Shape button () allows you to draw various geometric shapes and lines directly on the map for marking areas, creating boundaries, and visualizing spatial relationships.

2. How to Draw Shapes

2.1 Accessing Shape Tools

1. Click the Shape icon () on the Design Toolbar
2. A dropdown menu appears with all available shape types
3. Select the shape you want to draw
4. Cursor changes to crosshair indicating drawing mode

2.2 Available Shape Types

Draw Line

- Creates multi-segment connected lines
- Usage:
 - Click to place each vertex
 - Continue clicking to add segments
 - Double-click to finish
 - ESC to cancel

Circle

- Creates circular areas with defined radius
- Usage:
 - Single click to place center
 - Circle created with default size
 - Automatically exits drawing mode

Rectangle

- Creates rectangular areas

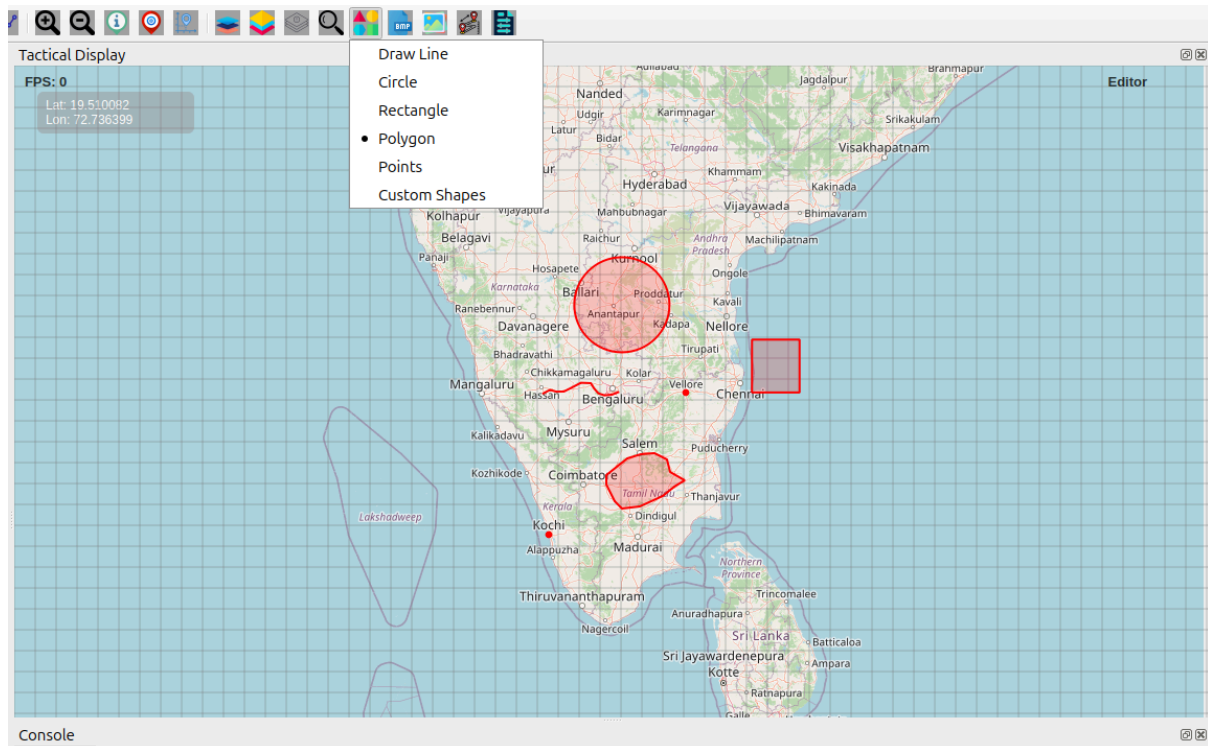
- Usage:
 - Single click to place rectangle
 - Default size rectangle created
 - Automatically exits drawing mode

Polygon 🏠

- Creates multi-sided closed shapes
- Usage:
 - Click to place each vertex
 - Continue clicking to add sides
 - Double-click to close and finish
 - ESC to cancel

Points •

- Creates individual point markers
- Usage:
 - Click to place each point
 - Each click creates a new point
 - Automatically stays in drawing mode



3. Shape Management & Editing

3.1 Selecting Shapes

- Switch to Move Mode (hand icon)
- Click on any shape to select it
- Selected shapes show selection outline

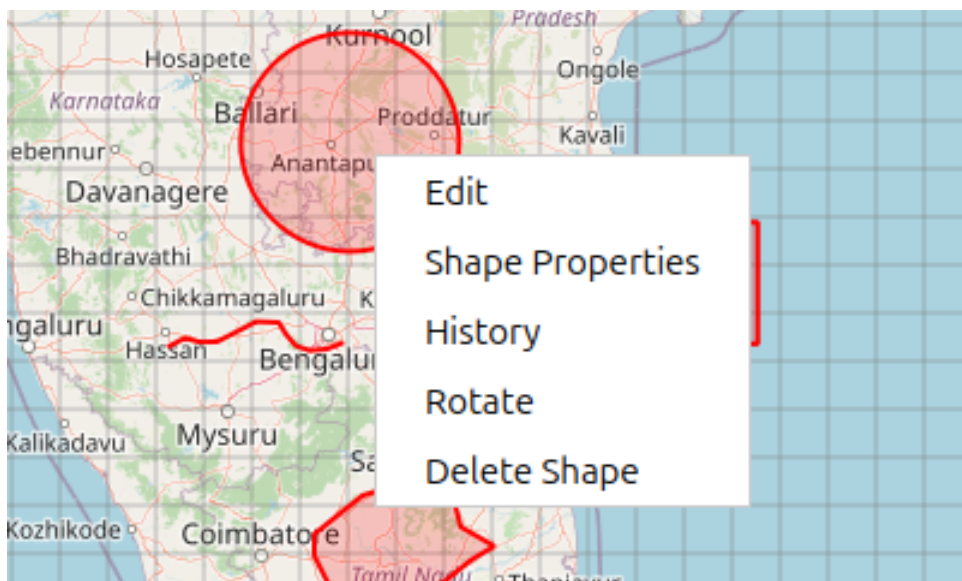
3.2 Moving Shapes

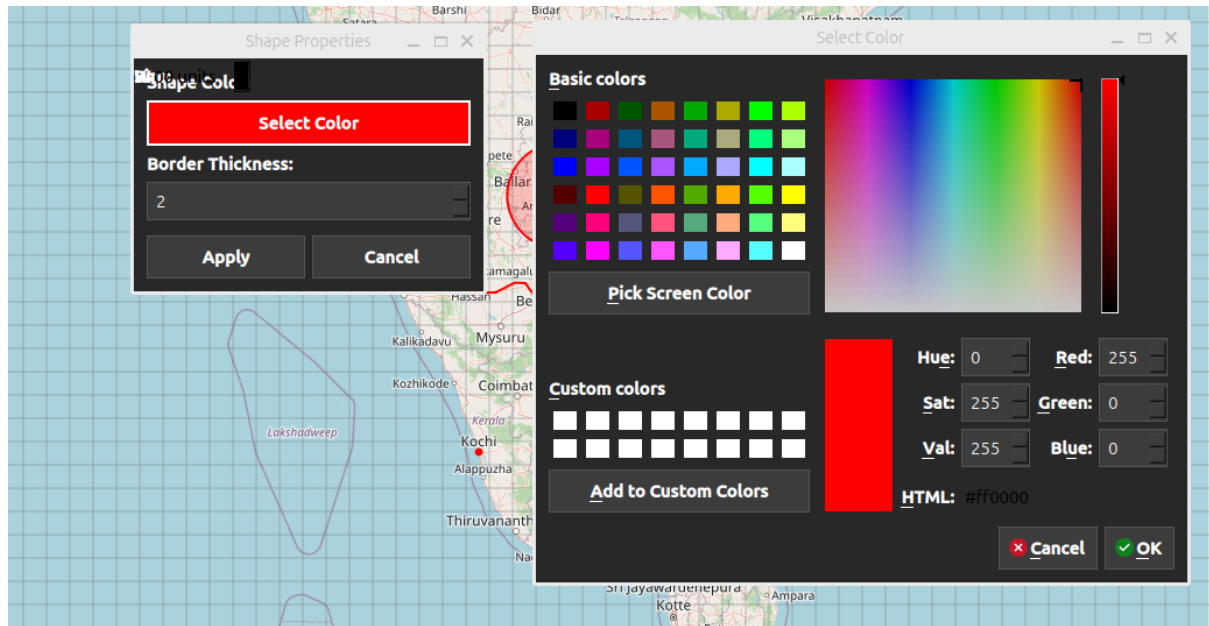
- Select shape in Move Mode
- Click and drag to reposition
- Real-time coordinate updates

3.3 Editing Shapes (Right-Click Menu)

Right-click any shape for options:

- Edit: Enter edit mode with resize handles
- Shape Properties: Change color and border thickness
- History: View movement history with green outlines
- Rotate: Activate rotation handles
- Delete Shape: Remove permanently





3.4 Resizing Shapes

1. Right-click shape → "Edit"
2. Resize handles appear at corners/edges
3. Drag handles to resize
4. Click outside to exit edit mode

3.5 Rotation

1. Right-click shape → "Rotate"
2. Rotation handle appears
3. Drag handle to rotate shape
4. Visual rotation guide displayed

Bitmap Features User Guide

1. Bitmap Icon Overview

The Bitmap button () allows you to place preset icons and custom images on the map for visual markers, points of interest, and custom annotations.

2. How to Use Bitmaps

2.1. Accessing Bitmap Tools

1. Click the Bitmap icon () on the Design Toolbar
2. A dropdown menu appears with preset bitmap options
3. Select the bitmap type you want to place
4. Cursor changes to crosshair indicating placement mode

2.2. Available Preset Bitmaps

Hospital

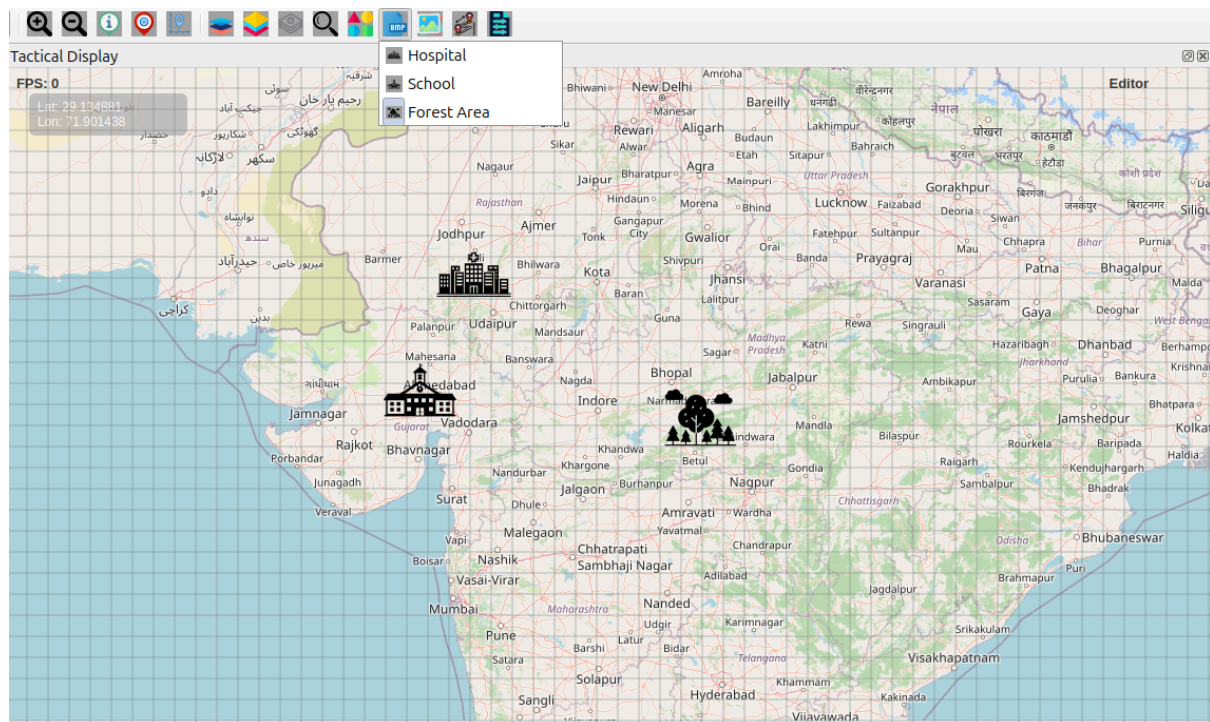
- Medical facilities and health services
- Emergency medical locations
- Health infrastructure markers

School

- Educational institutions
- Learning centers
- Academic facilities

Forest Area

- Wooded regions
- Natural reserves
- Environmental zones
- Park and recreation areas



3. Placing Bitmaps on Map

3.1. Placement Process

1. Select bitmap type from dropdown menu
2. Click on map at desired location
3. Bitmap automatically appears at clicked position
4. Returns to Move Mode automatically after placement

4. Bitmap Management & Editing

4.1. Selecting Bitmaps

- Switch to Move Mode (hand icon)
- Click directly on any bitmap to select it
- Selected bitmaps can be moved and edited

4.2. Moving Bitmaps

- Select bitmap in Move Mode
- Click and drag to reposition
- Real-time coordinate updates during movement

4.3. Editing Bitmaps (Right-Click Menu)

Right-click any bitmap for options:

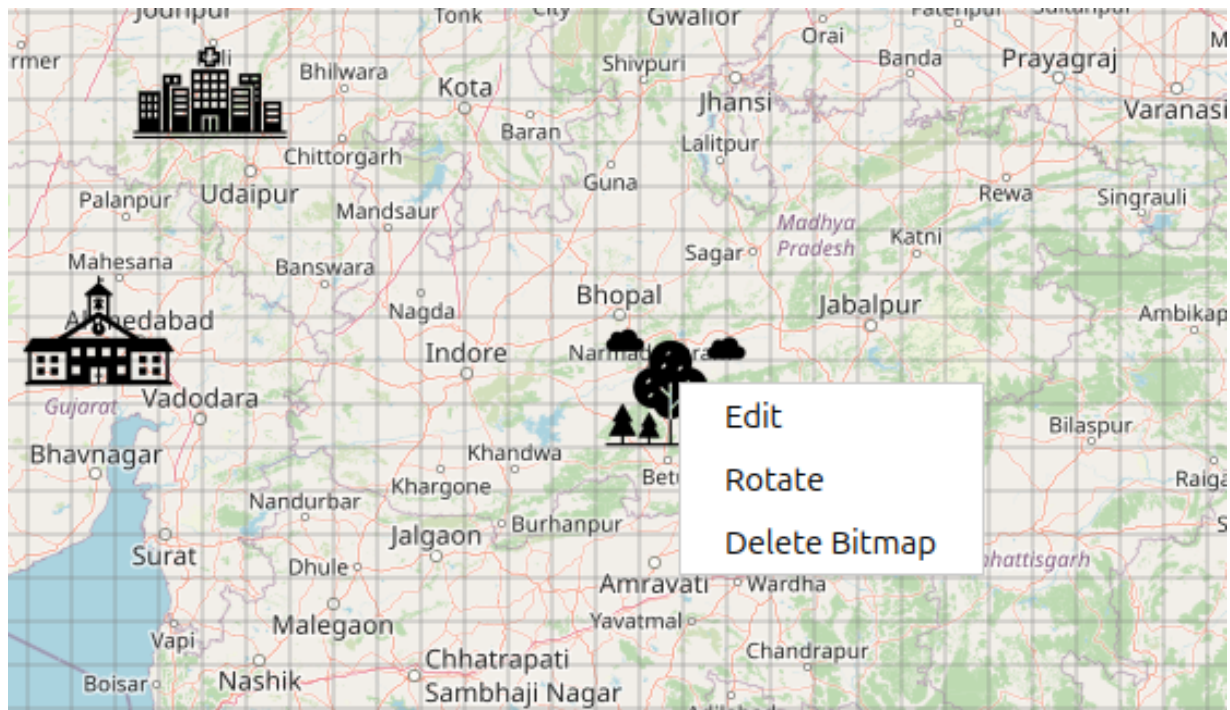
- Edit: Enter edit mode with resize handles
- Rotate: Activate rotation handles for orientation
- Delete Bitmap: Remove permanently from map

4.4. Resizing Bitmaps

1. Right-click bitmap → "Edit"
2. Resize handles appear at corners
3. Drag handles to adjust size
4. Maintains aspect ratio during resizing
5. Click outside to exit edit mode

4.5. Rotation

1. Right-click bitmap → "Rotate"
2. Rotation handle appears (yellow circle)
3. Drag handle to rotate bitmap
4. Visual rotation guide displayed
5. Smooth rotation with angle feedback



Select Bitmap Image Features User Guide

1. Select Bitmap Image Icon Overview

The Select Bitmap Image button () allows you to upload and place your own custom images on the map, providing complete flexibility for marking locations with company logos, specialized icons, or any visual markers you need.

2. How to Use Select Bitmap Image

2.1. Accessing the Feature

1. Click the Select Bitmap Image icon () on the Design Toolbar
2. File selection dialog opens immediately
3. Navigate to and select your image file
4. Click "Open" to place the image

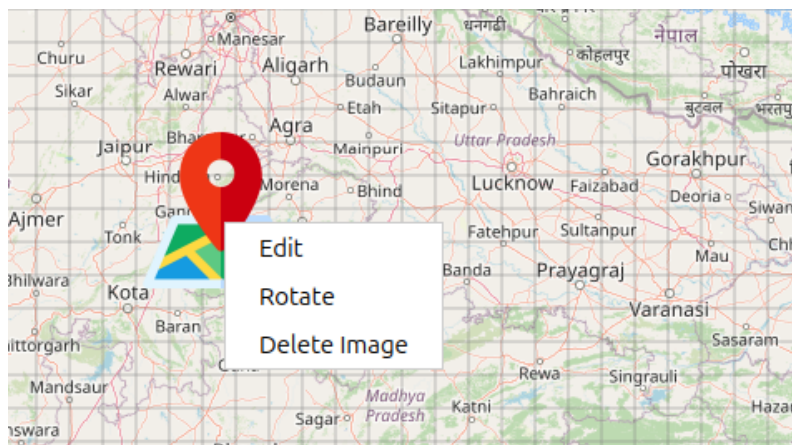
2.2. Supported File Formats

- PNG (.png) - Recommended (supports transparency)
- JPG/JPEG (.jpg, .jpeg) - Standard photo format

3. Placing Custom Images

3.1. Automatic Placement

- Image is automatically centered on your current map view
- No need to click on map - placement is instant
- Perfect for precise positioning using map navigation first



4. Managing Custom Images

4.1. Selecting Custom Images

- Switch to Move Mode (hand icon)
- Click directly on your custom image to select it
- Selected images show as active for editing

4.2. Moving Custom Images

- Select image in Move Mode
- Click and drag to reposition anywhere on map
- Real-time coordinate tracking during movement

4.3. Editing Custom Images (Right-Click Menu)

Right-click any custom image for options:

- Edit: Enter edit mode with resize handles
- Rotate: Activate rotation handles for orientation adjustment
- Delete Image: Remove permanently from map

4.4. Resizing Custom Images

1. Right-click image → "Edit"
2. Four resize handles appear at corners
3. Drag any handle to adjust size proportionally
4. Maintains original aspect ratio automatically
5. Click outside image to exit edit mode

4.5. Rotation

1. Right-click image → "Rotate"
2. Rotation handle appears (yellow circle)
3. Drag handle to rotate image to desired angle
4. Smooth rotation with visual feedback
5. Perfect for aligning images with features or roads

Measure Distance Features User Guide

1. Measure Distance Icon Overview

The Measure Distance button () allows you to accurately measure distances between points on the map with multiple calculation methods and unit options. Perfect for planning, analysis, and operational measurements.

2. How to Use Measure Distance

2.1. Starting Measurement

1. Click the Measure Distance icon () on the Design Toolbar
2. Cursor changes to crosshair indicating measurement mode
3. Measurement dialog automatically opens
4. You're now ready to measure

2.2. Measurement Process

1. Click on map to set start point
2. Continue clicking to add measurement points
3. Segments automatically connect between points
4. Real-time distance updates in dialog
5. Double-click or press ESC to finish

3. Measurement Dialog Features

3.1. Segments Display

- List of all measurement legs
- Formatted columns for easy reading:
 - X coordinate (longitude)
 - Y coordinate (latitude)
 - Distance for each segment
- Monospaced font for perfect alignment

3.2. Total Distance

- Running total of all segments
- Automatic unit conversion
- Read-only display for quick reference

4. Measurement Options

4.1. Calculation Methods

Ellipsoidal (Recommended - Default)

- High accuracy for long distances
- Uses WGS84 ellipsoid model
- Accounts for Earth's curvature
- Best for: Large areas, geographic measurements

Cartesian

- Faster calculation
- Flat-plane geometry
- Good for small areas
- Best for: Local measurements, quick estimates

4.2. Unit Selection

- Meters (default) - Base unit
- Kilometers - For longer distances
- Feet - Imperial system
- Miles - Road distances
- Degrees - Geographic coordinate distances

